INFRASTRUCTURE INVESTMENT CHARACTERISTICS

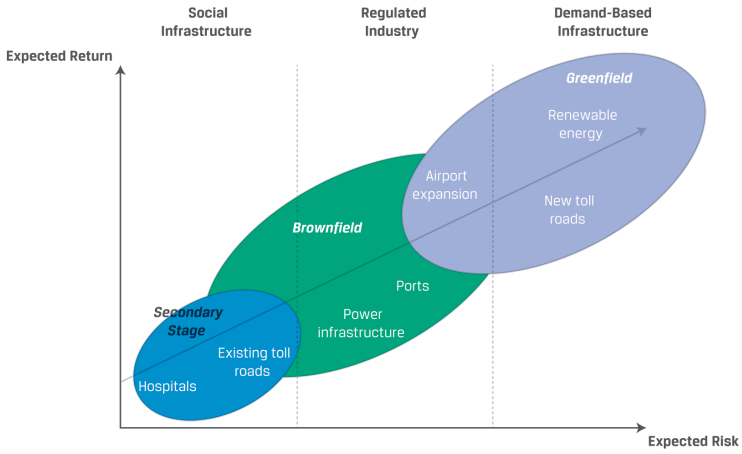
Learning Outcome

- explain the investment characteristics of infrastructure investments

The type of the underlying infrastructure investment, its stage of development, its geographic location, and the way the investment is structured define the expected risk and returns, as Exhibit 8 depicts.

Exhibit 8:

Infrastructure Investment Characteristics



▶ Expand / collapse Extended Description

In the three groupings of the infrastructure development cycle—greenfield, operational secondary-stage, and brownfield—the operational secondary-stage assets with an existing track record of generating steady, bond-like cash flows possess the lowest risk and offer the lowest return to the investors. Brownfield investments, redevelopment of existing infrastructure, are incrementally riskier, and greenfield projects are the riskiest. Exhibit 9 shows the historical risk and return statistics for infrastructure assets compared to other real asset categories.

Exhibit 9:

Historical Annualized Return and Standard Deviation of Return for Infrastructure and Real Assets, March 2008–June 2021

Annualized	Average return	Standard deviation	Coefficient of variation
Preqin, infrastructure	8.57%	0.07	0.82
Preqin, natural resources	3.29%	0.09	2.66
Preqin, real estate	3.43%	0.10	2.86
S&P Infrastructure Total Return	3.52%	0.19	5.52
MSCI US REIT Total Return	7.77%	0.26	3.38

Additionally, the type of infrastructure investment is material in determining the risk and return. While investments in basic social services infrastructure or existing regulated industries typically involve less risk and lower expected return, demand-based infrastructure projects often build on projections of future economic growth and increased usage demands and are riskier.

In developing market economies, where infrastructure investments are needed to underpin economic, social, and societal growth and benefit from increased wealth created by the economy, risks are considerable, but the returns—over long time horizons—are considerable. Greenfield infrastructure projects in developing countries offer exceptional return opportunities over very long time horizons. Exhibit 10 shows the risk profiles of various infrastructure projects.

Exhibit 10:

Private Infrastructure Fund Illustrative Target Returns

High-Risk Profile	Medium-Risk Profile	Low-Risk Profile
Greenfield projects without guarantees of demand upon completion—e.g., variable electricity prices, uncertain traffic on roads and through ports	Mostly brownfield assets (with some capital expenditure requirements) and some greenfield assets (with limited construction and demand risk)	Brownfield assets with mitigated risks—e.g., fully constructed with contracted/regulated revenues
Located in OECD countries and emerging markets	Located primarily in OECD countries	Located in the most stable OECD countries
High weighting to capital appreciation	Mix of yield and capital appreciation	High weighting to current yield
Target equity returns of 14%+	**Target equity returns of 10%–12%**	**Target equity returns of 6%–8%**

Note: Target equity returns are net of fees.

Source: Cambridge Associates, "Digging In: Assessing the Private Infrastructure Opportunity Today," Research Note (April 2017).

Most infrastructure funds gravitate toward the medium- and low-risk profiles, generating an average long-term annual return around 10%. As for other alternative investments, less liquid forms of direct equity ownership investments tend to offer the highest expected return with the greatest risk, while publicly traded forms of debt offer the lowest potential returns. Assets backed by stable long-term concession arrangements provide the most stable returns.

Infrastructure Diversification Benefits

Infrastructure investors expect the assets to primarily generate stable long-term cash flows that also adjust for economic growth and inflation and secondarily expect capital appreciation, depending on the type and timing of their investment. Because infrastructure investments typically support services that face inelastic demand and/or benefit from high barriers to entry, generate steady cash returns, and have a longer life cycle, equity investments in infrastructure offer lower correlation with public market equities and the broader economy. Effectively, infrastructure investments provide an income stream, increase portfolio diversification by adding an asset class with typically low correlation with other public investments, provide some protection for changes in GDP growth, and offer some protection against inflation. Exhibit 11 summarizes the diversification benefits of infrastructure investments and shows the low correlation between infrastructure and other market returns. It is notable that public and private infrastructure returns exhibit low correlations.

Exhibit 11:

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Correlation	S&P 500 Total Return	S&P Infrastructure Total Return	MSCI US REIT Total Return	MSCI World Total Return
Preqin, infrastructure	0.12	0.14	0.33	0.08
Preqin, natural resources	0.68	0.67	0.61	0.68
Preqin, real estate	0.51	0.39	0.49	0.46

Source: Annualized quarterly returns of Private Capital Quarterly Index rebased to 31 December 2007, provided by Preqin.

Most institutional investors consider infrastructure investments to balance public equity holdings because infrastructure has proven to be relatively resilient to swings in the equity markets. Given the stable underlying nature of cash flows, infrastructure debt tends to experience lower default rates and higher recovery than similar fixed-income instruments, and it is less subject to fluctuation over the economic cycle.

Additionally, infrastructure may better match the longer-term liability structure of certain investors, such as pension funds, superannuation schemes, and life insurance companies. It also suits the longer-term horizon of sovereign wealth funds, which tend to make the largest allocations to this asset class—around 5%–6% of total assets under management, according to Preqin. Long-term correlation benefits also result from the fact that most infrastructure assets have a link to inflation through regulation, concession agreements, or other fee contracts whose rates rise to or above the rate of inflation.

Question Set

1. True or false: Greenfield infrastructure projects in developing countries offer exceptional return opportunities over very long time horizons.

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Solution:

True. In developing market economies, where infrastructure investments are needed to underpin economic, social, and societal growth and benefit from increased wealth created by the economy, risks are considerable but the returns —over long time horizons—are also considerable.

2. Which of these types of infrastructure investment has the highest expected risk?

A. Regulated industry
B. Social infrastructure
C. Demand-based infrastructure

Hide Solution

Solution:

The correct answer is C. Demand-based infrastructure investments have the highest expected risk of the three. Social infrastructure has the lowest.

3. Which of the following has the highest weighting to capital appreciation?

A. Greenfield assets with limited construction and demand risk
B. Fully constructed brownfield assets with contracted revenues
C. Greenfield assets without guarantees of demand upon completion

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roads and through ports—have a high weighting to capital appreciation. Greenfield assets with limited construction and demand risk have a mix of yield and capital appreciation. Brownfield assets with mitigated risks (e.g., fully constructed with contracted/regulated revenues) that are located in the most stable OECD countries have a high weighting to current yield.

4. Which of the following is true regarding infrastructure investments?

A. Infrastructure investments typically generate volatile cash returns.
B. Infrastructure investments typically support services that face inelastic demand and/or benefit from high barriers to entry.
C. While public infrastructure returns have low correlation with market returns, private infrastructure returns have high correlation with market returns.

Hide Solution

Solution:

The correct answer is B. Because infrastructure investments typically support services that face inelastic demand and/or benefit from high barriers to entry, generate steady cash returns, and have a longer life cycle, equity investments in infrastructure offer lower correlation to public market equities and the broader economy. It is notable that public and private infrastructure returns exhibit low correlations.

OPTIONAL UNTESTED SEGMENT

UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Program Curriculum on Bloomberg.

{BI INFRA} BLOOMBERG INTELLIGENCE INFRASTRUCTURE

The Bloomberg BI INFRA function allows you to view infrastructure investments.

> "The BI homepage allows you to search or browse for dashboards, and is designed to forefront the information most relevant to you via your customized favorites and suggested content, so the data and analytics on the industries and topics you follow are easily accessible. The screen is divided into a control area, the navigation sidebar, and research and analysis views that allow you to display related coverage, research, and contributor data, based on your selections from the sidebar. BI INFRA is the Infrastructure Dashboard." -Bloomberg

To explore this function, type BI INFRA into the Bloomberg command line and press Enter or <GO>.

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